# REMARKS ON THREE SHARPNESS QUESTIONS FOR COMPOSITION OPERATORS ON SPACES OF DIRICHLET SERIES

ØISTEIN SØVIK

ABSTRACT. We answer three questions on composition operators $C_\varphi$ on Hilbert spaces of Dirichlet series. First, if $\varphi(s) = c_0 s + c_1 + \sum_{m\geq 2} c_m m^{-s}$ has characteristic $c_0 \geq 1$, then $a_N(C_\varphi) \geq (2N)^{-\operatorname{Re} c_1}$ on $\mathscr{H}^2$, so the logarithmic factor in the lower bound of Queffélec and Queffélec can be removed. Second, for the critical symbols $\varphi(s) = c_1 + \sum_{j=1}^{d} c_j q_j^{-s}$ with independent $q_j$ and $2 \leq d < \infty$, the approximation numbers satisfy $a_n(C_\varphi) \asymp n^{-(d-1)/2}$; this removes the logarithmic gap in a theorem of Queffélec and Seip. Third, we confirm a conjecture of Bayart and Brevig: if $\varphi$ is a Dirichlet polynomial of degree one or two with unrestricted range in $d \geq 3$ variables, then $C_\varphi$ maps $\mathscr{D}_\alpha$ boundedly into $\mathscr{D}_\beta$ whenever $d\beta + 9/8 \geq 2^\alpha$, so the exponent $(2^\alpha - 1)/d$ is not optimal.

## 1. INTRODUCTION

Let $\mathscr{H}^2$ denote the Hilbert space of Dirichlet series $f(s) = \sum_{n\geq 1} a_n n^{-s}$ with

$$\|f\|_{\mathscr{H}^2}^2 = \sum_{n=1}^{\infty} |a_n|^2 < \infty.$$

For $\alpha \geq 0$, let $\mathscr{D}_\alpha$ be the Hilbert space of Dirichlet series with

$$\|f\|_{\mathscr{D}_\alpha}^2 = \sum_{n=1}^{\infty} \frac{|a_n|^2}{[d(n)]^\alpha} < \infty,$$

where $d(n)$ is the number of divisors of $n$, so that $\mathscr{D}_0 = \mathscr{H}^2$. Write $\mathbb{C}_\theta = \{s : \operatorname{Re} s > \theta\}$. Gordon and Hedenmalm [4] showed that an analytic map $\varphi\colon \mathbb{C}_{1/2} \to \mathbb{C}_{1/2}$ generates a bounded composition operator $C_\varphi f = f \circ \varphi$ on $\mathscr{H}^2$ if and only if $\varphi$ belongs to the class $\mathscr{G}$ of functions

$$\varphi(s) = c_0 s + \sum_{n=1}^{\infty} c_n n^{-s} = c_0 s + \varphi_0(s),$$

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where the characteristic $c_0$ is a nonnegative integer, $\varphi_0$ converges uniformly in $\mathbb{C}_\varepsilon$ for every $\varepsilon > 0$, and $\varphi_0(\mathbb{C}_0) \subset \mathbb{C}_{1/2}$ if $c_0 = 0$, while $\varphi_0 \equiv 0$ or $\varphi_0(\mathbb{C}_0) \subset \mathbb{C}_0$ if $c_0 \geq 1$.

The approximation numbers of a bounded operator $T$ are

$$a_n(T) = \inf\{\|T - R\| : \operatorname{rank} R < n\}, \qquad n \geq 1.$$

They measure how fast $T$ can be approximated by finite rank operators, and $T$ belongs to the Schatten class $S_p$ if and only if $(a_n(T)) \in \ell^p$. Queffélec and Seip [6] initiated the study of $a_n(C_\varphi)$ on $\mathscr{H}^2$, Bayart, Queffélec and Seip [3] continued it on the spaces $\mathscr{H}^p$, and Bayart and Brevig [1] treated general characteristic-zero Dirichlet polynomials. The results are collected in Chapter 8 of the book of Queffélec and Queffélec [5]. Two of the estimates in that chapter leave a logarithmic gap.

**Positive characteristic.** Let $\varphi(s) = c_0 s + \psi(s)$ with $\psi(s) = \sum_{n\geq 1} c_n n^{-s}$, so that $c_1 = \psi(+\infty)$. If $c_0 \geq 1$ and $C_\varphi$ is compact on $\mathscr{H}^2$, then Queffélec and Queffélec [5, Theorem 8.8.1] prove that $a_n(C_\varphi) \geq \delta(n \log n)^{-c_1}$ for some $\delta > 0$, and they remark [5, p. 268] "The result of 1. is optimal, and that of 2. as well, up to the logarithmic factor". The symbol $\varphi(s) = c_0 s + c_1$ gives $a_n(C_\varphi) = n^{-\operatorname{Re} c_1}$, so the question is whether the logarithm is needed. The same lower bound, with $\operatorname{Re} c_1$ in the exponent, is [3, Theorem 1.2(b)].

**Theorem 1.** *Suppose that $\varphi(s) = c_0 s + c_1 + \sum_{m\geq 2} c_m m^{-s}$ belongs to $\mathscr{G}$ and that $c_0 \geq 1$. Then $\gamma = \operatorname{Re} c_1 \geq 0$, and*

$$a_N(C_\varphi) \geq (2N)^{-\gamma}$$

*for every integer $N \geq 1$. The power of $N$ is optimal.*

**The critical case in characteristic zero.** Let $1 \leq d < \infty$, let $q_1, \dots, q_d \geq 2$ be independent integers (so that the numbers $\log q_j$ are linearly independent over $\mathbb{Q}$), and consider

$$\varphi(s) = c_1 + \sum_{j=1}^{d} c_j q_j^{-s}, \qquad c_j \neq 0, \qquad K(\varphi) = \operatorname{Re} c_1 - \sum_{j=1}^{d} |c_j| = \frac{1}{2}. \tag{1}$$

By Kronecker's theorem, $\varphi \in \mathscr{G}$, and the condition $K(\varphi) = 1/2$ means that $\varphi$ touches the boundary line $\operatorname{Re} s = 1/2$. Queffélec and Seip [6, Theorem 1.3(a)] proved that

$$n^{-(d-1)/2} \ll a_n(C_\varphi) \ll n^{-(d-1)/2} (\log n)^{(d-1)/2}; \tag{2}$$

see also [5, Theorem 8.9.3], after which it is noted [5, p. 272]: "One obtains, in particular, up to a logarithmic factor: $a_n(C_\varphi) \approx n^{-(d-1)/2}$". For $d = 1$ both sides of (2) are of constant order, and $C_\varphi$ is not compact. Bayart and Brevig [1, Theorem 16 and Corollary 17] extended (2), with the same logarithmic factor, to more general symbols.

**Theorem 2.** *Suppose that $\varphi$ is as in (1) with $2 \le d < \infty$. Then*

$$a_n(C_\varphi) \asymp n^{-(d-1)/2}, \qquad n \ge 1,$$

*with constants depending only on $\varphi$.*

The lower bound is that of Queffélec and Seip [6, Section 8.2], so only the upper bound is new.

**Target exponents for quadratic symbols.** A Dirichlet polynomial $\varphi \in \mathscr{G}$ of characteristic zero has a Bohr lift $\Phi$, a polynomial in $d = \dim(\varphi)$ complex variables, and $\deg(\varphi)$ is the degree of $\Phi$; see Section 2. Following Bayart and Brevig [2], $\varphi$ has unrestricted range if $\varphi(\mathbb{C}_0) \not\subset \mathbb{C}_{1/2+\delta}$ for every $\delta > 0$. Bayart and Brevig [2, Theorem 17(i)] proved that if $\alpha > 0$, $\varphi$ has unrestricted range, $\dim(\varphi) = d \ge 2$ and $\deg(\varphi) \in \{1, 2\}$, then $C_\varphi$ maps $\mathscr{D}_\alpha$ boundedly into $\mathscr{D}_{(2^\alpha-1)/d}$. They then write [2, Remark after Theorem 21]:

> It is natural to ask whether the space $\mathscr{D}_{(2^\alpha-1)/d}$ in Theorem 17 (i) is optimal. We found that this is not the case for linear symbols in Theorem 20. By Theorem 21, it is optimal if $\dim(\varphi) = 2$. For $\dim(\varphi) \ge 3$, we conjecture that $(2^\alpha - 1)/d$ is not optimal, but our results do not further substantiate this claim.

**Theorem 3.** *Suppose that $\alpha > 0$ and that $\varphi \in \mathscr{G}$ is a Dirichlet polynomial with unrestricted range, $d = \dim(\varphi) \ge 3$ and $\deg(\varphi) \in \{1, 2\}$. Then $C_\varphi \colon \mathscr{D}_\alpha \to \mathscr{D}_\beta$ is bounded for every $\beta \ge 0$ with*

$$d\beta + 1 + \frac{1}{8} \ge 2^\alpha,$$

*and compact if the inequality is strict. In particular, $C_\varphi$ maps $\mathscr{D}_\alpha$ boundedly into $\mathscr{D}_{\beta_*}$ with*

$$\beta_* = \max\left\{0, \frac{2^\alpha - 1 - 1/8}{d}\right\} < \frac{2^\alpha - 1}{d},$$

*and $C_\varphi \colon \mathscr{D}_\alpha \to \mathscr{D}_{(2^\alpha-1)/d}$ is compact.*

This confirms the conjecture, with a gain $1/8$ in the numerator that is uniform over all such symbols. The last statement strengthens [2, Corollary 18(i)], which gives compactness into $\mathscr{D}_{2^\alpha-1}$. In the other direction, Theorem 4 in Section 7 exhibits, for every $d \ge 3$, a quadratic symbol for which the exact threshold is $d\beta + 1 + (d-1)/4 \ge 2^\alpha$. So the best uniform gain lies between $1/8$ and $(d-1)/4$. For linear symbols, Bayart and Brevig [2, Theorem 20] found the exact threshold $d\beta + 1 + (d-1)/2 \ge 2^\alpha$.

To the best of our knowledge, Theorems 1 to 3 are new.

**Our approach.** Let us briefly describe the three arguments. For Theorem 1, the proof of Queffélec and Queffélec tests $C_\varphi$ on the span of $p_1^{-s}, \dots, p_N^{-s}$, where $p_k$ is the $k$th prime, and the factor $\log N$ is the size of $p_N/N$. Primes are used because the leading terms $j^{-c_1}(j^{c_0})^{-s}$ of $C_\varphi j^{-s}$ must not interfere

with the other terms. We observe that the integers $N < j \le 2N$ already have this property: no $k^{c_0}$ is a proper multiple of $j^{c_0}$ when $1/2 < k/j < 2$.

For Theorem 2, we write $C_\varphi^* C_\varphi$ as the embedding of $\mathscr{H}^2$ into $L^2(\mu)$, where $\mu$ is the distribution of $\varphi$ on the boundary. Near the contact point, $\mathscr{H}^2$ behaves like the Hardy space of a half-plane, that is, like the space of Laplace transforms $\int_0^\infty F(\xi) e^{-w\xi}\,\mathrm{d}\xi$, and the frequency $\xi$ corresponds to $\log n$. The measure $\mu$ lives on a parabolic region $|\operatorname{Im} w| \lesssim (\operatorname{Re} w)^{1/2}$. A function of frequency $\xi \in [\Xi, 2\Xi)$ is only visible to $\mu$ for $\operatorname{Re} w \lesssim 1/\Xi$, and there $\mu$ sees a box of height $\Xi^{-1/2}$; such functions therefore have about $\Xi^{1/2}$ degrees of freedom. Queffélec and Seip truncate all frequencies at once, with a single Blaschke product, and this costs the logarithm. We split the frequencies into dyadic blocks, estimate the approximation numbers of each block in operator norm, and allot dimensions to the blocks so that the errors form a geometric series. In contrast to [6], the construction is in the frequency variable rather than in the half-plane.

For Theorem 3, we follow the proof of [2, Theorem 17(i)] and only improve its last step. Bayart and Brevig estimate the pullback of a Carleson square $Q(\tau, \varepsilon)$ by localizing every radius $1 - |z_j|$ to $O(\varepsilon)$ and one angle to an interval of length $O(\varepsilon)$, which gives $\varepsilon^{d\beta+1}$. The remaining angles are also constrained, since the real part of $\Phi$ on the torus must be at most $1/2 + \varepsilon$. This gains a factor $\varepsilon^{1/m}$ if the real part vanishes to order $m$ in some direction inside the level set of the imaginary part. The gain is lost only if $\Phi$ touches the line $\operatorname{Re} s = 1/2$ along a curve, as $\Phi(z) = 1/2 + (1 - z_1 z_2)$ does in dimension two. In dimension $d \ge 3$ we show that a quadratic $\Phi$ cannot do this, even to order 8. The key inequality is $\operatorname{Re} \Phi(-1, \dots, -1) \ge 1/2$, and the last step is a finite Taylor computation, which we have also verified by computer in exact arithmetic.

**Organization.** The present paper is organized into seven additional sections. In Section 2 we collect the facts on approximation numbers and on Bohr lifts that we use. Section 3 contains the proof of Theorem 1. In Section 4 we reduce Theorem 2 to a weighted embedding of Laplace transforms, and in Section 5 we estimate this embedding block by block and prove Theorem 2. Section 6 contains the proof of Theorem 3, and Section 7 the family of examples. Section 8 contains concluding remarks and two problems.

## 2. Preliminaries

**Approximation numbers.** Let $T$ be a bounded operator between Hilbert spaces. We use four elementary facts.

- (A1) $a_n(ATB) \le \|A\| a_n(T) \|B\|$ (the ideal property).
- (A2) If $E$ is a subspace of dimension $n$ and $\|Tf\| \ge c\|f\|$ for every $f \in E$, then $a_n(T) \ge c$. Indeed, if $\operatorname{rank} R < n$, then $R$ annihilates a unit vector $f \in E$, and $\|T - R\| \ge \|Tf\| \ge c$.

(A3) $a_{m+k+1}(S+T) \le a_{m+1}(S) + a_{k+1}(T)$ for $m, k \ge 0$, since the sum of an operator of rank at most $m$ and one of rank at most $k$ has rank at most $m+k$. By induction, $a_{1+\sum_i m_i}(\sum_i T_i) \le \sum_i a_{m_i+1}(T_i)$ for finitely many operators.

(A4) $a_n(T)^2 = a_n(T^*T) = a_n(TT^*)$, and for a positive compact operator the approximation numbers are the eigenvalues in decreasing order.

If $\|Sf\| \le C\|Tf\|$ for every $f$, then $S = VT$ for an operator $V$ with $\|V\| \le C$, defined by $V(Tf) = Sf$ on the range of $T$ and extended by continuity and by zero on the orthogonal complement. Hence (A1) gives

$$a_n(S) \le C a_n(T). \tag{3}$$

**The Bohr lift.** Let $p_1 < p_2 < \cdots$ be the primes. A Dirichlet polynomial $f(s) = \sum_n a_n n^{-s}$ is identified with the polynomial $\mathscr{B}f(z) = \sum_n a_n z^{\kappa(n)}$, where $n = \prod_j p_j^{\kappa_j}$; then $\|f\|^2_{\mathscr{H}^2}$ is the integral of $|\mathscr{B}f|^2$ over the polytorus with respect to Haar measure. If $\varphi \in \mathscr{G}$ is a Dirichlet polynomial of characteristic zero, then $\Phi = \mathscr{B}\varphi$ is a polynomial in finitely many variables $z_1, \dots, z_d$, and $\mathscr{B}(f \circ \varphi) = f \circ \Phi$. For the symbols (1) it is more convenient to use the variables $z_j = q_j^{-s}$; by independence of the $q_j$, distinct multi-indices $\kappa$ give distinct integers $q^\kappa = \prod_j q_j^{\kappa_j}$, which is all that will be used.

For $\beta > 0$ let $\mathrm{d}\tilde{m}_\beta(z) = \beta(1-|z|^2)^{\beta-1}\,\mathrm{d}m_1(z)$ on $\mathbb{D}$, where $m_1$ is Lebesgue measure normalized by $m_1(\mathbb{D}) = 1$, and let $\tilde{\nu}_\beta$ be the product measure on $\mathbb{D}^d$. For $\beta = 0$ we let $\tilde{\nu}_0$ be Haar measure on $\mathbb{T}^d$. The norm of $\mathscr{D}_\beta$ is computed with a different product measure $\nu_\beta$ [2, (6)], but on polynomials in $d$ variables the two give comparable norms, and either may be used in the criteria below [2, Section 3.3]. For $\tau \in \mathbb{R}$ and $\varepsilon > 0$ let

$$Q(\tau, \varepsilon) = [1/2, 1/2+\varepsilon] \times [\tau - \varepsilon/2, \tau + \varepsilon/2]$$

be the Carleson square, and let $\mu_{\beta,\varphi}(E) = \nu_\beta(\{z : \Phi(z) \in E\})$ be the pullback measure on $\mathbb{C}_{1/2}$, with $z \in \mathbb{D}^d$ if $\beta > 0$ and $z \in \mathbb{T}^d$ if $\beta = 0$.

The following two results of Bayart and Brevig are our tools for Theorems 3 and 4. Let $\varphi \in \mathscr{G}$ be a Dirichlet polynomial of characteristic zero with Bohr lift $\Phi$ in $d$ variables, and let $\alpha, \beta \ge 0$.

(B1) By [2, Lemma 11], the operator $C_\varphi \colon \mathscr{D}_\alpha \to \mathscr{D}_\beta$ is bounded if and only if $\mu_{\beta,\varphi}(Q(\tau,\varepsilon)) = O(\varepsilon^{2^\alpha})$ uniformly in $\tau$, and compact if and only if $\mu_{\beta,\varphi}(Q(\tau,\varepsilon)) = o(\varepsilon^{2^\alpha})$ as $\varepsilon \to 0^+$, uniformly in $\tau$.

(B2) By [2, Corollary 12], the following holds. Suppose that for every $w \in \mathbb{T}^d$ with $\operatorname{Re}\Phi(w) = 1/2$ there are a neighborhood $U_w$ of $w$, a constant $C_w$ and $\kappa_w \ge 2^\alpha$ such that $\tilde{\nu}_\beta(\{z \in U_w : \Phi(z) \in Q(\tau,\varepsilon)\}) \le C_w \varepsilon^{\kappa_w}$ for every $\tau \in \mathbb{R}$ and $\varepsilon > 0$. Then $C_\varphi \colon \mathscr{D}_\alpha \to \mathscr{D}_\beta$ is bounded. If moreover $\kappa_w > 2^\alpha$ for every such $w$, then it is compact.

## 3. Positive characteristic

We first record how $C_\varphi$ acts on the basis $e_j(s) = j^{-s}$ when $c_0 \geq 1$. Write $\varphi(s) = c_0 s + c_1 + \psi_1(s)$ with $\psi_1(s) = \sum_{m\geq 2} c_m m^{-s}$. Then

$$C_\varphi e_j(s) = j^{-c_1}(j^{c_0})^{-s} \exp\bigl(-(\log j)\psi_1(s)\bigr).$$

The exponential is a Dirichlet series $1 + \sum_{m\geq 2} \alpha_{j,m} m^{-s}$ with constant term one, since $\psi_1$ has no constant term. Hence

$$C_\varphi e_j = j^{-c_1} e_{j^{c_0}} + j^{-c_1} \sum_{m\geq 2} \alpha_{j,m} e_{j^{c_0} m}. \tag{4}$$

This formal computation is justified in [5, Lemma 8.8.2 and its proof], where it is used with $j$ prime. The leading term $j^{-c_1} e_{j^{c_0}}$ has norm $j^{-\operatorname{Re} c_1}$, and the point of the proof below is to choose $N$ integers $j$ whose leading frequencies $j^{c_0}$ are not hit by the remaining terms of any other $C_\varphi e_k$.

*Proof of Theorem 1.* By the definition of $\mathscr{G}$, either $\psi = c_1 + \psi_1 \equiv 0$ or $\operatorname{Re}\psi(s) > 0$ for $s \in \mathbb{C}_0$. Letting $s \to +\infty$ along the real axis, we get $\gamma = \operatorname{Re} c_1 \geq 0$.

Let $I_N = \{N+1, \dots, 2N\}$, let $E_N$ be the span of $\{e_j : j \in I_N\}$, and let $P$ be the orthogonal projection onto the span of $\{e_{k^{c_0}} : k \in I_N\}$. Suppose that $j, k \in I_N$ and $j^{c_0} m = k^{c_0}$ for an integer $m \geq 1$. Then $(k/j)^{c_0} = m$ is an integer. A rational number whose positive integer power is an integer is itself an integer, so $k/j$ is an integer. Since $1/2 < k/j < 2$, we get $k = j$ and $m = 1$. By (4), the only term of $C_\varphi e_j$ which survives the projection $P$ is the leading one, and

$$PC_\varphi\Bigl(\sum_{j\in I_N} b_j e_j\Bigr) = \sum_{j\in I_N} b_j j^{-c_1} e_{j^{c_0}}.$$

The vectors $e_{j^{c_0}}$, $j \in I_N$, are orthonormal. Thus, for $f = \sum_{j\in I_N} b_j e_j$ in $E_N$,

$$\|C_\varphi f\|^2 \geq \|PC_\varphi f\|^2 = \sum_{j\in I_N} |b_j|^2 j^{-2\gamma} \geq (2N)^{-2\gamma}\|f\|^2.$$

Since $\dim E_N = N$, (A2) gives $a_N(C_\varphi) \geq (2N)^{-\gamma}$.

For optimality, let $\varphi(s) = c_0 s + c_1$. Then $C_\varphi e_j = j^{-c_1} e_{j^{c_0}}$, so $C_\varphi$ maps an orthonormal basis to an orthogonal family with norms $j^{-\gamma}$, which are decreasing in $j$. Hence $a_N(C_\varphi) = N^{-\gamma}$; see also [5, p. 270]. □

The proof does not use compactness, and for $\gamma = 0$ it gives $a_N(C_\varphi) = \|C_\varphi\| = 1$. In the proof of [5, Theorem 8.8.1] the test space is spanned by $p_1^{-s}, \dots, p_N^{-s}$, and the bound $p_N^{-\gamma} \gg (N \log N)^{-\gamma}$ is where the logarithm enters. The first proof given there, which applies to $c_0 = 1$, combines Bayart's formula $\lambda_n = n^{-c_1}$ for the eigenvalues with Weyl's inequality, and the logarithm is the price of passing from eigenvalues to approximation numbers.

## 4. The critical case: a model operator

In this section and the next, $\varphi$ is as in (1) with $2 \le d < \infty$. Constants $C, c$ depend only on $\varphi$ and may change from line to line. We write

$$r_j = |c_j|, \qquad r = \sum_{j=1}^{d} r_j, \qquad R = 3r, \qquad s_0 = \frac{1}{2} + i \operatorname{Im} c_1,$$

and $W(z) = \sum_j r_j(1 - z_j)$ for $z \in \mathbb{T}^d$. Let $m_d$ be normalized Haar measure on $\mathbb{T}^d$, and let $\mu$ be the image of $m_d$ under $W$. Thus $\mu$ is a probability measure on the disc $\{|w - r| \le r\}$, which touches the imaginary axis only at 0. We write $w = x + iy$ throughout.

**Lemma 1.** *For every $f \in \mathscr{H}^2$, $\|C_\varphi f\|^2 = \int |f(s_0 + w)|^2 \, \mathrm{d}\mu(w)$.*

*Proof.* Let $\Phi(z) = c_1 + \sum_j c_j z_j$, so that $\varphi(s) = \Phi(q_1^{-s}, \dots, q_d^{-s})$. For a Dirichlet polynomial $f$, the function $f \circ \Phi$ is entire on $\mathbb{C}^d$, with power series $\sum_\kappa b_\kappa z^\kappa$, and $f \circ \varphi = \sum_\kappa b_\kappa (q^\kappa)^{-s}$. The integers $q^\kappa$ are distinct, so $\|f \circ \varphi\|^2 = \sum_\kappa |b_\kappa|^2 = \int_{\mathbb{T}^d} |f \circ \Phi|^2 \, \mathrm{d}m_d$. Put $u_j = -c_j z_j / r_j$. Then $\Phi(z) = c_1 - \sum_j r_j u_j = s_0 + W(u)$, because $\operatorname{Re} c_1 - r = 1/2$, and the map $z \mapsto u$ preserves $m_d$. This proves the identity for Dirichlet polynomials. For general $f$, take Dirichlet polynomials $f_k \to f$ in $\mathscr{H}^2$. By the identity for polynomials, $(Jf_k)$ is a Cauchy sequence in $L^2(\mu)$, where $Jf(w) = f(s_0 + w)$. Since $f_k \to f$ pointwise in $\mathbb{C}_{1/2}$ and $\mu$-almost every $w$ has $\operatorname{Re} w > 0$, its limit is $Jf$. Hence $\|Jf\|_{L^2(\mu)} = \lim_k \|C_\varphi f_k\| = \|C_\varphi f\|$. □

Hence $a_n(C_\varphi) = a_n(J)$, where $Jf(w) = f(s_0 + w)$ maps $\mathscr{H}^2$ into $L^2(\mu)$. Recall that the reproducing kernel of $\mathscr{H}^2$ at $s \in \mathbb{C}_{1/2}$ is $\zeta(\cdot + \bar{s})$. Near $s = 1$ we have $\zeta(s) \approx 1/(s-1)$, and $1/(w + \bar{w}')$ is the reproducing kernel of the Laplace transforms

$$\mathcal{L}F(w) = \int_0^\infty F(\xi) e^{-w\xi} \, \mathrm{d}\xi, \qquad F \in L^2(0, \infty),$$

since $\int_0^\infty e^{-(w+\bar{w}')\xi} \, \mathrm{d}\xi = 1/(w + \bar{w}')$. This suggests replacing $J$ by $\mathcal{L}$. We first describe $\mu$ near 0 by a simpler measure. Let $\eta$ be the measure on $\mathbb{C}_0$ given by

$$\mathrm{d}\eta(w) = x^{(d-3)/2} e^{-y^2/x} \mathbf{1}_{(0,R)}(x) \, \mathrm{d}x \, \mathrm{d}y.$$

The factor $e^{-y^2/x}$ expresses that $\mu$ lives on the parabolic region $|y| \lesssim \sqrt{x}$, and the power $x^{(d-3)/2}$ that a box of size $x$ has $\mu$-measure about $x^{(d+1)/2}$: each of the $d$ angles is $O(\sqrt{x})$, and one of them is pinned down to precision $x$ by $y$.

**Lemma 2.** *For every $F \in L^2(0, \infty)$, $\int |\mathcal{L}F|^2 \, \mathrm{d}\mu \le C \int |\mathcal{L}F|^2 \, \mathrm{d}\eta$.*

*Proof.* Put $g = \mathcal{L}F$, which is analytic in $\mathbb{C}_0$. For $w$ in $\mathbb{C}_0$ let $D_w$ be the disc of radius $x_w/2$ centred at $w$, where $x_w = \operatorname{Re} w$. Since $|g|^2$ is subharmonic,

$|g(w)|^2$ is at most its mean over $D_w$. Integrating against $\mu$ and using Fubini's theorem, we find that

$$\int |g|^2 \,\mathrm{d}\mu \le \int |g(v)|^2 h(v)\,\mathrm{d}x_v\,\mathrm{d}y_v, \qquad h(v) = \int \frac{4\mathbf{1}_{D_w}(v)}{\pi x_w^2}\,\mathrm{d}\mu(w),$$

where $v = x_v + iy_v$. It remains to show that $h(v) \le C x_v^{(d-3)/2} e^{-y_v^2/x_v}$ for $0 < x_v < R$, and $h(v) = 0$ for $x_v \ge R$.

If $v \in D_w$, then $2x_v/3 < x_w < 2x_v$ and $|v - w| < x_v$. Hence

$$h(v) \le \frac{9}{\pi x_v^2}\,\mu\big(\{w : x_w < 2x_v,\ |y_w - y_v| < x_v\}\big).$$

Since $x_w \le 2r$ on the support of $\mu$, we get $h(v) = 0$ when $x_v \ge 3r = R$.

Now write $w = W(e^{i\theta})$ with $\theta \in [-\pi, \pi)^d$. Then

$$x_w = \sum_j r_j(1 - \cos\theta_j) \ge \frac{2\min_j r_j}{\pi^2}|\theta|^2, \qquad |y_w| = \Big|\sum_j r_j \sin\theta_j\Big| \le r|\theta|.$$

Hence $|y_w| \le C\sqrt{x_w}$ on the support of $\mu$, and $h(v) = 0$ unless $|y_v| \le C\sqrt{x_v}$; on that set $e^{-y_v^2/x_v}$ is bounded below. Let $t = x_v$. The condition $x_w < 2t$ forces $|\theta_j| \le k\sqrt{t}$ for every $j$, with $k$ depending only on $\varphi$. Suppose first that $k\sqrt{t} \le \pi/3$, and fix $\theta_2, \dots, \theta_d$. On $|\theta_1| \le k\sqrt{t}$ the derivative of $\theta_1 \mapsto y_w = -\sum_j r_j \sin\theta_j$ has modulus at least $r_1/2$, so the condition $|y_w - y_v| < t$ confines $\theta_1$ to an interval of length at most $4t/r_1$. By Fubini's theorem, $\mu(\{x_w < 2t,\ |y_w - y_v| < t\}) \le Ct \cdot t^{(d-1)/2}$, and so $h(v) \le Ct^{(d-3)/2}$. If $k\sqrt{t} > \pi/3$, then $t$ lies in a fixed compact subinterval of $(0, R)$, and $h(v) \le 9/(\pi t^2) \le C$. In both cases $h(v) \le C x_v^{(d-3)/2} e^{-y_v^2/x_v}$. □

The next lemma is the only estimate for $\mathcal{L}F$ that we need. It is Plancherel's theorem on vertical lines, with the Gaussian weight used to estimate tails in $y$.

**Lemma 3.** *Suppose that $F \in L^2(0,\infty)$ vanishes on $(0, \Xi)$, where $\Xi \ge 0$. Then for every $x > 0$ and $Y \ge 0$,*

$$\int_{|y|>Y} |\mathcal{L}F(x+iy)|^2 e^{-y^2/x}\,\mathrm{d}y \le 2\pi e^{-Y^2/x - 2x\Xi}\|F\|^2.$$

*In particular, $\int |\mathcal{L}F|^2\,\mathrm{d}\eta \le 2\pi\|F\|^2 \int_0^R x^{(d-3)/2} e^{-2x\Xi}\,\mathrm{d}x$, which is finite because $d \ge 2$.*

*Proof.* The function $y \mapsto \mathcal{L}F(x+iy)$ is the Fourier transform of $F(\xi)e^{-x\xi}$. By Plancherel's theorem,

$$\int_{\mathbb{R}} |\mathcal{L}F(x+iy)|^2\,\mathrm{d}y = 2\pi \int_\Xi^\infty |F(\xi)|^2 e^{-2x\xi}\,\mathrm{d}\xi \le 2\pi e^{-2x\Xi}\|F\|^2,$$

and on $|y| > Y$ the weight $e^{-y^2/x}$ is at most $e^{-Y^2/x}$. □

Let $L\colon L^2(0,\infty) \to L^2(\eta)$ and $M\colon L^2(0,\infty) \to L^2(\mu)$ both be given by $F \mapsto \mathcal{L}F$. By Lemma 3, $L$ is bounded. By Lemma 2, $\|MF\| \le C\|LF\|$, so $M$ is bounded and, by (3),

$$a_n(M) \le Ca_n(L). \tag{5}$$

The next lemma shows that $J$ and $M$ differ by an operator with tiny approximation numbers.

**Lemma 4.** *For every $\rho \in (0,1)$ there is $C_\rho$ such that $a_{2n-1}(C_\varphi)^2 \le a_n(M)^2 + C_\rho\rho^n$ for every $n \ge 1$.*

*Proof.* Let $g \in L^2(\mu)$ be bounded and supported in $\{x \ge \delta\}$ for some $\delta > 0$. Then $J^*g = \int \zeta(\cdot + \overline{s_0 + w'})g(w')\,\mathrm{d}\mu(w')$ and $M^*g(\xi) = \int e^{-\bar{w}'\xi}g(w')\,\mathrm{d}\mu(w')$. Since $s_0 + \bar{s}_0 = 1$, we get

$$JJ^*g(w) = \int \zeta(1+w+\bar{w}')g(w')\,\mathrm{d}\mu(w'), \qquad MM^*g(w) = \int \frac{g(w')}{w+\bar{w}'}\,\mathrm{d}\mu(w').$$

Let $E(u) = \zeta(u) - 1/(u-1)$, which is an entire function. Let $B$ be the operator on $L^2(\mu)$ with the bounded hermitian kernel $E(1 + w + \bar{w}')$. Since $\mu(\{x = 0\}) = 0$, functions $g$ as above are dense in $L^2(\mu)$, and hence $JJ^* = MM^* + B$.

We expand $E(u) = \sum_k e_k(u-u_0)^k$ about $u_0 = 1 + 2r$. By Cauchy's estimates, for every $\rho \in (0,1)$ there is $C_\rho$ with $|e_k| \le C_\rho(\rho/2r)^k$. Since $u - u_0 = (w - r) + \overline{(w' - r)}$, the binomial theorem gives

$$E(1 + w + \bar{w}') = \sum_{i,l\ge 0} e_{i+l}\binom{i+l}{i}(w-r)^i\,\overline{(w'-r)^l}.$$

The part of this sum with $i, l < N$ is a kernel of rank at most $N$. On the support of $\mu$ we have $|w - r| \le r$, and $\mu$ is a probability measure, so the remaining part has operator norm at most

$$\sum_{k\ge N}|e_k|\sum_{i=0}^{k}\binom{k}{i}r^k = \sum_{k\ge N}|e_k|(2r)^k \le \frac{C_\rho\rho^N}{1-\rho}.$$

Hence $a_{N+1}(B) \le C'_\rho\rho^N$. By (A4) and (A3), $a_{2n-1}(J)^2 = a_{2n-1}(JJ^*) \le a_n(MM^*) + a_n(B) \le a_n(M)^2 + C'_\rho\rho^{n-1}$. Finally $a_n(C_\varphi) = a_n(J)$ by Lemma 1. □

In view of Lemma 4 and (5), Theorem 2 is reduced to the estimate $a_n(L) \le Cn^{-(d-1)/2}$ for the weighted Laplace embedding $L$. This is the content of the next section.

## 5. The critical case: dyadic frequency blocks

For an interval $I \subset (0,\infty)$ let $P_I$ be the orthogonal projection of $L^2(0,\infty)$ onto $L^2(I)$. The functions $F$ supported in $[\Xi, 2\Xi)$ correspond to Dirichlet series with frequencies $\log n$ in $[\Xi, 2\Xi)$. By Lemma 3, $\mathcal{L}F$ is negligible unless $x \lesssim 1/\Xi$, and the Gaussian weight is negligible unless $|y| \lesssim \Xi^{-1/2}$. On such

a box, $\mathcal{L}F$ is well approximated by a polynomial of degree about $\Xi^{1/2}$ times $e^{-w\Xi}$. This is the content of part (i) of the next lemma.

**Lemma 5.** *The following estimates hold.*

(i) *If* $\Xi \geq 1$, *then* $a_{n+1}(LP_{[\Xi,2\Xi)}) \leq C\Xi^{-(d-1)/4}e^{-n/(8e\sqrt{\Xi})}$ *for every* $n \geq 0$.
(ii) $a_{n+1}(LP_{[0,1)}) \leq Ce^{-cn}$ *for every* $n \geq 0$.
(iii) *If* $\Xi \geq 1$, *then* $\|LP_{[\Xi,\infty)}\| \leq C\Xi^{-(d-1)/4}$.

*Proof.* We have

$$\int_0^\infty x^{(d-3)/2}e^{-2x\Xi}\,\mathrm{d}x = C\Xi^{-(d-1)/2},$$

so part (iii) follows from Lemma 3. Part (iii) also gives (i) when $n < 4e\sqrt{\Xi}$, since then $e^{-n/(8e\sqrt{\Xi})} \geq e^{-1/2}$.

For (i) with $n \geq 4e\sqrt{\Xi}$, put $M_0 = n/(4e\sqrt{\Xi}) \geq 1$ and

$$\Omega = \{w : x \leq M_0/\Xi,\ |y| \leq M_0/\sqrt{\Xi}\}.$$

Let $F$ be supported in $[\Xi, 2\Xi)$. Expanding $e^{-w\tau}$ in a power series, we get

$$\mathcal{L}F(w) = e^{-w\Xi}\int_0^\Xi F(\Xi+\tau)e^{-w\tau}\,\mathrm{d}\tau = e^{-w\Xi}\sum_{j\geq 0}\frac{(-w)^j m_j}{j!},$$

where $m_j = \int_0^\Xi F(\Xi+\tau)\tau^j\,\mathrm{d}\tau$, and the Cauchy–Schwarz inequality gives $|m_j| \leq \Xi^{j+1/2}\|F\|$. Define the operator $R_n$ of rank at most $n$ by

$$R_nF(w) = \mathbf{1}_\Omega(w)e^{-w\Xi}\sum_{j<n}\frac{(-w)^j m_j}{j!}.$$

We split $\|(L-R_n)F\|^2_{L^2(\eta)}$ into three parts.

On $\Omega$ we have $|w|\Xi \leq 2M_0\sqrt{\Xi} = n/(2e)$. Since $j! \geq (j/e)^j$, each term with $j \geq n$ satisfies $(|w|\Xi)^j/j! \leq 2^{-j}$, and hence

$$|\mathcal{L}F(w) - R_nF(w)| \leq \sqrt{\Xi}\|F\|\sum_{j\geq n}\frac{(|w|\Xi)^j}{j!} \leq 2^{1-n}\sqrt{\Xi}\|F\|.$$

Moreover $\eta(\Omega) \leq \sqrt{\pi}\int_0^{M_0/\Xi} x^{(d-2)/2}\,\mathrm{d}x \leq C(M_0/\Xi)^{d/2}$. So the contribution of $\Omega$ is at most $C\Xi^{1-d/2}M_0^{d/2}4^{-n}\|F\|^2$. Since $n = 4eM_0\sqrt{\Xi} \geq 5M_0 + 5\sqrt{\Xi}$, this is at most $C\Xi^{-(d-1)/2}e^{-M_0}\|F\|^2$.

Off $\Omega$, $R_nF = 0$. By Lemma 3 with $Y = 0$, the contribution of $\{x > M_0/\Xi\}$ is at most

$$2\pi\|F\|^2\int_{M_0/\Xi}^\infty x^{(d-3)/2}e^{-2x\Xi}\,\mathrm{d}x \leq C\Xi^{-(d-1)/2}e^{-M_0}\|F\|^2.$$

By Lemma 3 with $Y = M_0/\sqrt{\Xi}$, the contribution of the set where $x \leq M_0/\Xi$ and $|y| > Y$ is at most

$$2\pi\|F\|^2\int_0^\infty x^{(d-3)/2}e^{-2x\Xi - M_0^2/(\Xi x)}\,\mathrm{d}x \leq 2\pi e^{-2M_0}\|F\|^2\int_0^\infty x^{(d-3)/2}e^{-x\Xi}\,\mathrm{d}x,$$

because $x\Xi + M_0^2/(\Xi x) \geq 2M_0$ by the inequality between the arithmetic and geometric means. The last integral is $C\Xi^{-(d-1)/2}$. Adding the three contributions, we find that $a_{n+1}(LP_{[\Xi,2\Xi)})^2 \leq C\Xi^{-(d-1)/2}e^{-M_0}$, which is (i).

For (ii), let $F$ be supported in $[0,1)$. Then $\mathcal{L}F(w) = \sum_j (-w)^j m_j/j!$ with $|m_j| \leq \|F\|$. Let $n > 2eR$, put $Y = n/(2e) - R$, and let $R_nF$ be the truncation of this series to $j < n$, multiplied by $\mathbf{1}_{\{|y|\leq Y\}}$; for $n \leq 2eR$ the bound follows from the boundedness of $L$. On the support of $\eta$ and $|y| \leq Y$ we have $|w| \leq n/(2e)$, so the error there is at most $2^{1-n}\|F\|$, and $\eta$ is a finite measure. By Lemma 3 with $\Xi = 0$, the rest contributes at most $Ce^{-Y^2/R}\|F\|^2$. □

The rank $n$ in part (i) enters only through $n/\sqrt{\Xi}$: a block of frequencies $[\Xi, 2\Xi)$ has about $\sqrt{\Xi}$ effective dimensions and norm $\Xi^{-(d-1)/4}$. We now allot dimensions to the blocks $[2^k, 2^{k+1})$.

*Proof of Theorem 2.* The lower bound is [6, Theorem 1.3(a)]. For the upper bound, it suffices by Lemma 4 and (5) to show that $a_N(L) \leq CN^{-(d-1)/2}$.

Fix an integer $K \geq 1$ and put $b = 1 + (d-1)(\log 2)/4$. Let $n_k = \lceil 8eb(K-k)2^{k/2} \rceil$ for $0 \leq k \leq K$, and let $n_{-1} = \lceil K(d-1)(\log 2)/(4c) \rceil$ with $c$ from part (ii) of Lemma 5. We write $L$ as the sum of $LP_{[0,1)}$, of $LP_{[2^k,2^{k+1})}$ for $0 \leq k \leq K$, and of $LP_{[2^{K+1},\infty)}$. Let $N = 1 + n_{-1} + \sum_{k=0}^K n_k$. By (A3) and Lemma 5,

$$a_N(L) \leq Ce^{-cn_{-1}} + C\sum_{k=0}^{K} 2^{-k(d-1)/4}e^{-b(K-k)} + C2^{-(K+1)(d-1)/4}.$$

The first term is at most $C2^{-K(d-1)/4}$. By the choice of $b$,

$$2^{-k(d-1)/4}e^{-b(K-k)} = 2^{-K(d-1)/4}e^{-(K-k)}.$$

Summing the geometric series, we get

$$a_N(L) \leq C2^{-K(d-1)/4}.$$

On the other hand, $N \leq C(K + \sum_{k\leq K}(K-k+1)2^{k/2}) \leq C_1 2^{K/2}$. Given $n \geq C_1 2^{1/2}$, choose the largest $K$ with $C_1 2^{K/2} \leq n$. Then $2^{(K+1)/2} > n/C_1$, and since $a_n(L)$ is nonincreasing,

$$a_n(L) \leq C2^{-K(d-1)/4} \leq Cn^{-(d-1)/2}.$$

By Lemma 4 and (5), $a_{2n-1}(C_\varphi) \leq Cn^{-(d-1)/2}$, and the theorem follows, since $a_n(C_\varphi)$ is nonincreasing. □

## 6. Quadratic symbols

In this section we prove Theorem 3. We need two lemmas. The first is an elementary sublevel set estimate of van der Corput type.

**Lemma 6.** *Let $m \geq 1$, let $I \subset \mathbb{R}$ be an interval, and let $g \in C^m(I)$ be real with $|g^{(m)}| \geq c > 0$ on $I$. Then $|\{t \in I : |g(t)| \leq \varepsilon\}| \leq C_m(\varepsilon/c)^{1/m}$ for every $\varepsilon > 0$, where $C_1 = 2$ and $C_m = 2 + 2C_{m-1}$.*

*Proof.* For $m = 1$, $g$ is strictly monotone with $|g'| \geq c$, so the set is an interval of length at most $2\varepsilon/c$. Let $m \geq 2$ and $\delta > 0$. By the case $m = 1$ applied to $g^{(m-1)}$, the set where $|g^{(m-1)}| < c\delta$ is an interval of length at most $2\delta$. Its complement in $I$ consists of at most two intervals on which $|g^{(m-1)}| \geq c\delta$. By induction, the set in question has measure at most $2\delta + 2C_{m-1}(\varepsilon/(c\delta))^{1/(m-1)}$. Now choose $\delta = (\varepsilon/c)^{1/m}$. □

The second lemma is a finite Taylor computation for two-variable quadratic polynomials of a special form, which will appear in the proof as the restriction of $\Phi$ to two of its variables. Let $a_1, a_2 > 0$ and $x_1, x_2, \omega \in \mathbb{R}$, and consider

$$p(z_1, z_2) = \sum_{j=1}^{2} \Big[ a_j(1 - z_j) + \Big( \frac{a_j}{2} - a_j^2 + i x_j \Big)(1 - z_j)^2 \Big] + (-2a_1a_2 + i\omega)(1 - z_1)(1 - z_2). \tag{6}$$

On the torus put $z_1 = e^{ia_2(u+v)}$ and $z_2 = e^{ia_1(u-v)}$, and let $G(u, v) = \operatorname{Re} p$ and $H(u, v) = \operatorname{Im} p$. Expanding $1 - e^{it} = -it + t^2/2 + it^3/6 + \cdots$, we find that

$$G(u, v) = 4a_1^2a_2^2u^2 + O(|u|^3 + |v|^3), \qquad H(u, v) = -2a_1a_2u + O(u^2 + v^2).$$

So $u$ is the direction in which $H$ changes, and $v$ is the direction of the level curve $H = 0$ at the origin. Since $\partial_u^2 G(0) \neq 0$, the implicit function theorem gives a real-analytic curve $u = A(v)$ with $A(0) = 0$ and $\partial_u G(A(v), v) = 0$. Put $f_1(v) = G(A(v), v)$ and $f_3(v) = H(A(v), v)$, and write $[f]_k$ for the coefficient of $v^k$ in the Taylor series of $f$ at 0.

**Lemma 7.** *If $a_1 + a_2 < 1$, then $[f_1]_3$, $[f_1]_4$, $[f_3]_2$, $[f_3]_3$ and $[f_3]_4$ do not all vanish.*

*Proof.* Put $S = a_1 + a_2$, so that $0 < S < 1$. Solving $\partial_u G(A(v), v) = 0$ for the Taylor coefficients of $A$ order by order and substituting, we get

$$[f_1]_3 = a_2^3x_1 - a_1^3x_2 + \frac{a_1a_2(a_1 - a_2)}{2}\omega,$$
$$8a_1a_2[f_3]_2 = a_2^3(6 - 8a_1)x_1 + a_1^3(6 - 8a_2)x_2 + a_1a_2(8a_1a_2 - S)\omega.$$

As a linear system in $(x_1, x_2)$, the equations $[f_1]_3 = [f_3]_2 = 0$ have determinant $-a_1^2a_2^2(2S - 3)/2 \neq 0$, so they force

$$x_1 = \frac{a_1(2a_1a_2 + a_1 + 2a_2^2 - 2a_2)}{2a_2^2(2S - 3)}\omega, \qquad x_2 = \frac{a_2(2a_1^2 + 2a_1a_2 - 2a_1 + a_2)}{2a_1^2(2S - 3)}\omega.$$

With these values,

$$[f_1]_4 = -\frac{S(S^3\omega^2 + V)}{4(2S - 3)^2}, \qquad V = a_1a_2(2S - 3)^2(a_1a_2S - a_1^2 - a_2^2 + a_1a_2),$$

so $[f_1]_4 = 0$ forces $S^3\omega^2 + V = 0$. Modulo this relation,

$$[f_3]_3 = \frac{(a_1 - a_2)P}{4S}, \qquad 4P = -3S^2(1-S) - (9-5S)(a_1-a_2)^2 < 0.$$

Hence $[f_3]_3 = 0$ forces $a_1 = a_2 = a$, with $0 < a < 1/2$. Then the relation $S^3\omega^2 + V = 0$ reads $\omega^2 = -a(2a-1)(4a-3)^2/8$, so $\omega \neq 0$, and modulo this relation

$$[f_3]_4 = \frac{3a^2(2a-1)}{8(4a-3)}\omega \neq 0.$$

This proves the lemma. □

All displayed identities in the proof are identities between rational functions of $a_1, a_2, x_1, x_2, \omega$, obtained by expanding $G$ and $H$ to order five. An identity which holds "modulo" a relation means that the numerator of the difference of its two sides is divisible by $S^3\omega^2 + V$ (or, when $a_1 = a_2 = a$, by $\omega^2 + a(2a-1)(4a-3)^2/8$) as a polynomial in $\omega$. All denominators which occur are products of powers of $a_1$, $a_2$, $S$, $2S-3$ and $4a-3$, so they do not vanish when $0 < S < 1$. The identities are routine but lengthy, and we have verified them in exact rational arithmetic; see the code availability statement at the end of the paper.

We are now ready to prove Theorem 3. The first two steps, the normal form and the radial localization, are those of Bayart and Brevig [2, proof of Theorem 17(i)]. The third step is new.

*Proof of Theorem 3.* Fix $\beta \geq 0$. By (B2), it suffices to verify its hypothesis with $\kappa_w = d\beta + 1 + 1/8$ at every $w \in \mathbb{T}^d$ with $\operatorname{Re}\Phi(w) = 1/2$. The statement about $\beta = (2^\alpha - 1)/d$ then follows, since $d\beta + 9/8 > 2^\alpha$ for this $\beta$.

*Normal form.* After a rotation of each variable, which preserves $\tilde{\nu}_\beta$, we may take $w = \mathbf{1} = (1, \dots, 1)$. We write $\Phi(\mathbf{1}) = 1/2 + i\tau_0$ and

$$P(z) = \Phi(z) - \Phi(\mathbf{1}) = \sum_{j=1}^{d} a_j(1-z_j) + \sum_{j=1}^{d} b_j(1-z_j)^2 + \sum_{j<k} c_{j,k}(1-z_j)(1-z_k).$$

Then $\operatorname{Re} P \geq 0$ on $\overline{\mathbb{D}}^d$, and every variable occurs in $P$. As shown at the beginning of [2, proof of Theorem 17(i)], using [2, Lemma 19], $a_j > 0$ for every $j$.

*Radial localization.* Write $z_j = (1-\rho_j)e^{i\theta_j}$, and let $G(\theta) = \operatorname{Re} P(e^{i\theta})$ and $H(\theta) = \operatorname{Im} P(e^{i\theta})$, where $e^{i\theta} = (e^{i\theta_1}, \dots, e^{i\theta_d})$. As in [2], Taylor's formula in $\rho$ gives $\operatorname{Re} P(z) = \sum_j \rho_j F_j(\rho, \theta) + G(\theta)$, with $F_j$ smooth and $F_j(0,0) = a_j$, while $G \geq 0$ and $|P(z) - P(e^{i\theta})| \leq L\sum_j \rho_j$. Hence there are $\rho_0 > 0$, a neighborhood $V$ of $0 \in \mathbb{R}^d$ (to be shrunk below) and $c_0 \geq 1$ such that, for $U_w = \{z : \rho_j < \rho_0,\ \theta \in V\}$, every $z \in U_w$ with $\Phi(z) \in Q(\tau, \varepsilon)$ satisfies

$$\rho_j \leq c_0\varepsilon, \qquad G(\theta) \leq \varepsilon, \qquad |H(\theta) - (\tau - \tau_0)| \leq c_0\varepsilon. \tag{7}$$

Bayart and Brevig use only the first and the last conditions in (7). The last one confines one angle to an interval of length $O(\varepsilon)$, and this gives the exponent $d\beta + 1$. We use the second condition as well.

*The angular estimate.* We claim that there are a neighborhood $V$ of 0 and a constant $C$ such that for all $t \in \mathbb{R}$ and $0 < \varepsilon \le 1$,

$$\big|\{\theta \in V : G(\theta) \le \varepsilon,\ |H(\theta) - t| \le \varepsilon\}\big| \le C\varepsilon^{1+1/8}. \tag{8}$$

Note that $H(\theta) = -\sum_j a_j\theta_j + O(|\theta|^2)$. It suffices to find local coordinates $(h, v, \sigma) \in \mathbb{R}\times\mathbb{R}\times\mathbb{R}^{d-2}$ near 0, with $h = H(\theta)$, and an integer $2 \le m \le 8$ such that $\partial_v^m(G\circ\theta)(0) \ne 0$, where $\theta = \theta(h, v, \sigma)$. Indeed, then $|\partial_v^m(G\circ\theta)| \ge c > 0$ on a small box $B$, and we let $V$ be the image of $B$. For fixed $(h, \sigma)$, Lemma 6 and $G \ge 0$ bound the $v$-section of the set in (8) by $C_m(\varepsilon/c)^{1/m}$, and $h$ ranges over an interval of length $2\varepsilon$. The Jacobian of the change of coordinates is bounded, so the set has measure at most $C\varepsilon^{1+1/m} \le C\varepsilon^{1+1/8}$.

Since $G \ge 0$ and $G(0) = 0$, the quadratic Taylor form $Q$ of $G$ at 0 is positive semidefinite. Expanding $1 - e^{i\theta} = -i\theta + \theta^2/2 + i\theta^3/6 + \cdots$, we find that the quadratic Taylor form of $G$ is

$$Q(\theta) = \sum_j \Big(\frac{a_j}{2} - \operatorname{Re} b_j\Big)\theta_j^2 - \sum_{j<k} \operatorname{Re} c_{j,k}\theta_j\theta_k$$

and that its cubic Taylor form is

$$\sum_j \operatorname{Im} b_j\theta_j^3 + \sum_{j<k} \frac{\operatorname{Im} c_{j,k}}{2}\theta_j\theta_k(\theta_j + \theta_k).$$

There are three cases.

(a) $Q(e) > 0$ for some $e$ with $\sum_j a_j e_j = 0$, that is, for some $e$ tangent to the level set of $H$. Take $h = H(\theta)$ and linear coordinates $(v, \sigma)$ on the space of $(\theta_2, \dots, \theta_d)$, chosen so that the curve $\{h = 0,\ \sigma = 0\}$ has tangent $e$ at 0. This is possible since $\partial_{\theta_1} H(0) = -a_1 \ne 0$ and $e \ne 0$ has a nonzero component outside the first coordinate. Since the gradient of $G$ vanishes at 0, we get $\partial_v^2(G\circ\theta)(0) = c'Q(e) \ne 0$ for some $c' > 0$, so $m = 2$.

(b) $Q = 0$. Then the cubic Taylor form of $G \ge 0$ vanishes as well, since it is odd. Comparing coefficients, $b_j = a_j/2$ and $c_{j,k} = 0$. Since $\operatorname{Re}(1 - e^{i\theta})^2 = -2\cos\theta(1 - \cos\theta)$, this gives $G(\theta) = \sum_j a_j(1 - \cos\theta_j)^2$. Take $h = H(\theta)$, $v = \theta_2$ and $\sigma = (\theta_3, \dots, \theta_d)$. On $h = 0$, $\sigma = 0$ we have $\theta_1 = O(v)$, so $G\circ\theta = c''v^4 + O(v^5)$ with $c'' \ge a_2/4$, and $m = 4$.

(c) $Q \ne 0$, and $Q$ vanishes on the hyperplane $\sum_j a_j\theta_j = 0$. A positive semidefinite form which vanishes on a hyperplane is a multiple of the square of the defining linear form, so $Q = \lambda(\sum_j a_j\theta_j)^2$ with $\lambda > 0$. Replacing $P$ by $\lambda P$, which preserves $\operatorname{Re} P \ge 0$ and replaces each $a_j$ by $\lambda a_j$, we may take $\lambda = 1$. Comparing with the formula for $Q$, we get

$$\operatorname{Re} b_j = \frac{a_j}{2} - a_j^2, \qquad \operatorname{Re} c_{j,k} = -2a_ja_k.$$

Let $S_d = \sum_j a_j$. At $z = (-1, \dots, -1)$, where $1 - z_j = 2$, this gives

$$0 \le \operatorname{Re} P(-1, \dots, -1) = 2\sum_j a_j + 4\sum_j \operatorname{Re} b_j + 4\sum_{j<k} \operatorname{Re} c_{j,k} = 4S_d(1 - S_d).$$

Hence $S_d \le 1$, and since $d \ge 3$ and all $a_j > 0$, we get $a_1 + a_2 < 1$. Restricting $P$ to $z_3 = \cdots = z_d = 1$ gives the polynomial (6), with $x_j = \operatorname{Im} b_j$ and $\omega = \operatorname{Im} c_{1,2}$. Take $h = H(\theta)$, $v = (\theta_1/a_2 - \theta_2/a_1)/2$ and $\sigma = (\theta_3, \dots, \theta_d)$; the Jacobian of $(h, v)$ with respect to $(\theta_1, \theta_2)$ at 0 is 1. On $\sigma = 0$, the variables $(u, v)$ of (6) are $u = (\theta_1/a_2 + \theta_2/a_1)/2$ and $v$. Thus the function $\Psi(v) = (G \circ \theta)(0, v, 0)$ equals $G(u(v), v)$, where $u(v)$ solves $H(u(v), v) = 0$, in the notation of Lemma 7. Taylor's formula in $u$ about the curve $u = A(v)$, where $\partial_u G$ vanishes, gives

$$G(u,v) = f_1(v) + (u - A(v))^2 \tilde{A}(u,v), \qquad H(u,v) = f_3(v) + (u - A(v))\tilde{B}(u,v),$$

with $\tilde{A}, \tilde{B}$ real-analytic, $\tilde{A}(0) = 4a_1^2 a_2^2$ and $\tilde{B}(0) = -2a_1 a_2$. On $H = 0$ we have $u(v) - A(v) = -f_3(v)/\tilde{B}$, and so

$$\Psi(v) = f_1(v) + f_3(v)^2 \, \frac{\tilde{A}}{\tilde{B}^2}(u(v), v).$$

Both terms are nonnegative near 0, since $f_1 \ge 0$ by $G \ge 0$. Hence their leading Taylor coefficients cannot cancel, and the order of vanishing of $\Psi$ at 0 is the smaller of the orders of $f_1$ and of $f_3^2$. If it exceeded 8, then $f_1$ would vanish to order at least 9 and $f_3$ to order at least 5, which contradicts Lemma 7. So $m \le 8$.

*Conclusion.* Let $\beta > 0$. In polar coordinates we have $\mathrm{d}\tilde{m}_\beta = \beta(1 - |z|^2)^{\beta-1}|z| \, \mathrm{d}|z| \, \mathrm{d}\theta/\pi$, so the set of $z \in \mathbb{D}$ with $1 - |z| \le c_0\varepsilon$ and argument in a set $E$ has measure at most $|E|(2c_0\varepsilon)^\beta$. By (7), Fubini's theorem and (8) with $c_0\varepsilon$ in place of $\varepsilon$,

$$\tilde{\nu}_\beta\big(\{z \in U_w : \Phi(z) \in Q(\tau, \varepsilon)\}\big) \le C\varepsilon^{d\beta + 1 + 1/8}$$

for $0 < \varepsilon \le 1/c_0$, uniformly in $\tau$. For larger $\varepsilon$ the bound holds after enlarging $C$. For $\beta = 0$ the same argument, with every $\rho_j = 0$ and Haar measure on $\mathbb{T}^d$, gives the bound $C\varepsilon^{1+1/8}$. This is the hypothesis of (B2) with $\kappa_w = d\beta + 1 + 1/8$, which proves boundedness when $\kappa_w \ge 2^\alpha$ and compactness when $\kappa_w > 2^\alpha$. □

The restriction $d \ge 3$ enters only in case (c), through $a_1 + a_2 < 1$. For $d = 2$ the equality case $a_1 = a_2 = 1/2$, $x_1 = x_2 = \omega = 0$ of (6) is $p = (1 - z_1 z_2)/2$. This is a multiple of the Bohr lift of $3/2 - (p_1 p_2)^{-s}$, for which $\Psi \equiv 0$: the symbol touches the line $\operatorname{Re} s = 1/2$ along the curve $z_1 z_2 = 1$, and the exponent $(2^\alpha - 1)/2$ is optimal [2, Theorem 21].

## 7. A family of quadratic symbols

The next result shows how much can be gained in the numerator for one natural family. The symbols are those of case (b) in the proof of Theorem 3, where the angular estimate gains $\varepsilon^{1/4}$ in each of the $d - 1$ directions along the level set of the imaginary part.

**Theorem 4.** *Suppose that $d \geq 3$ and that $p_1, \dots, p_d$ are distinct primes, and let*

$$\varphi_d(s) = \frac{1}{2} + \sum_{j=1}^{d} \Big[ (1 - p_j^{-s}) + \frac{1}{2}(1 - p_j^{-s})^2 \Big].$$

*Then $\varphi_d \in \mathscr{G}$ has unrestricted range, and for $\alpha > 0$ and $\beta \geq 0$ the operator $C_{\varphi_d} \colon \mathscr{D}_\alpha \to \mathscr{D}_\beta$ is bounded if and only if*

$$d\beta + 1 + \frac{d-1}{4} \geq 2^\alpha.$$

*It is compact if and only if the inequality is strict.*

*Proof.* Put $F(z) = (1 - z) + (1 - z)^2/2$, so that the Bohr lift of $\varphi_d$ is $\Phi(z) = 1/2 + \sum_j F(z_j)$. Direct expansion gives

$$\operatorname{Re} F(\varrho e^{i\theta}) = \frac{1 - \varrho^2}{2} + (1 - \varrho\cos\theta)^2. \tag{9}$$

Thus $\operatorname{Re}\Phi > 1/2$ on $\mathbb{D}^d$, so $\varphi_d \in \mathscr{G}$. Its range is unrestricted, since $\varphi_d(s) \to 1/2$ as $s \to 0^+$. On $\mathbb{T}^d$, equality $\operatorname{Re}\Phi = 1/2$ holds only at $\mathbf{1}$. We show that, for all sufficiently small $\varepsilon > 0$,

$$\mu_{\beta,\varphi_d}(Q(0,\varepsilon)) \geq c\varepsilon^{d\beta+1+(d-1)/4}, \qquad \sup_{\tau\in\mathbb{R}} \mu_{\beta,\varphi_d}(Q(\tau,\varepsilon)) \leq C\varepsilon^{d\beta+1+(d-1)/4}. \tag{10}$$

By (B1), this proves the theorem: the upper bound gives boundedness when $d\beta + 1 + (d-1)/4 \geq 2^\alpha$ and compactness when the inequality is strict, and the lower bound at the single center $\tau = 0$ rules out boundedness below the threshold and compactness at equality. Since $\nu_\beta$ and $\tilde{\nu}_\beta$ have comparable densities near $\mathbb{T}^d$, we may compute with $\tilde{\nu}_\beta$.

First let $\beta > 0$, and write $z_j = \varrho_j e^{i\theta_j}$ with $-\pi \leq \theta_j \leq \pi$. If $\Phi(z) \in Q(\tau,\varepsilon)$, then (9) gives $1 - \varrho_j \leq 1 - \varrho_j^2 \leq 2\varepsilon$ and $1 - \varrho_j\cos\theta_j \leq \sqrt{\varepsilon}$ for every $j$. For small $\varepsilon$ the latter forces $\cos\theta_j > 0$, hence $1 - \cos\theta_j \leq 1 - \varrho_j\cos\theta_j \leq \sqrt{\varepsilon}$, and so $|\theta_j| \leq C\varepsilon^{1/4}$. Moreover,

$$\operatorname{Im} F(\varrho e^{i\theta}) = -2\varrho\sin\theta + \frac{\varrho^2}{2}\sin 2\theta, \qquad \frac{\partial}{\partial\theta}\operatorname{Im} F(\varrho e^{i\theta}) = -2\varrho\cos\theta + \varrho^2\cos 2\theta,$$

which tends to $-1$ as $(\varrho,\theta) \to (1,0)$. Hence, for fixed radii and fixed $\theta_2, \dots, \theta_d$, the condition on $\operatorname{Im}\Phi$ confines $\theta_1$ to an interval of length at most $2\varepsilon$ when $\varepsilon$ is small. Each radial condition $1 - \varrho_j \leq 2\varepsilon$ has $\tilde{m}_\beta$-measure at most $(4\varepsilon)^\beta$, and each $\theta_j$ with $j \geq 2$ ranges over an interval of length $C\varepsilon^{1/4}$. By Fubini's theorem, this proves the upper bound in (10).

For the lower bound, fix a small $\delta > 0$ depending only on $d$, and restrict each radius to $1 - \delta\varepsilon \leq \varrho_j \leq 1$ and each $\theta_j$, $2 \leq j \leq d$, to $|\theta_j| \leq \delta\varepsilon^{1/4}$. Since $\operatorname{Im} F(\varrho e^{i\theta}) = -(2\varrho - \varrho^2)\theta + O(|\theta|^3)$, the implicit function theorem gives a unique small $\theta_1^* = \theta_1^*(\varrho, \theta_2, \dots, \theta_d)$ with $\operatorname{Im}\Phi = 0$, and $|\theta_1^*| \leq 2d\delta\varepsilon^{1/4}$. For

$|\theta_1 - \theta_1^*| \le \varepsilon/4$ we have $|\operatorname{Im}\Phi| \le \varepsilon/2$, and by (9),

$$\operatorname{Re}\Phi - \frac{1}{2} \le \sum_j \Big(\delta\varepsilon + (\delta\varepsilon + \theta_j^2/2)^2\Big) \le \varepsilon$$

if $\delta$ is small enough. So $\Phi(z) \in Q(0,\varepsilon)$ on a set of $\tilde\nu_\beta$-measure at least $c(\delta\varepsilon)^{d\beta}(\delta\varepsilon^{1/4})^{d-1}\varepsilon$. This proves the lower bound in (10). For $\beta = 0$ the pullback measure is computed with Haar measure on $\mathbb{T}^d$, and the same angular arguments, with every $\varrho_j = 1$, give (10) without the radial factor. □

## 8. Concluding remarks

**Where the logarithms came from.** In both theorems of Queffélec and Queffélec, the logarithm is not intrinsic to the operator. In Theorem 1 it comes from the size of the $N$th prime, and in Theorem 2 from a single Blaschke product truncation at the scale $1/2 + n^{-2}(\log n)^2$ in [6, Section 8.3]. In the proof of Theorem 2 it is essential that each block is estimated in operator norm. A Hilbert–Schmidt estimate of the block $[\Xi, 2\Xi)$ loses a factor $\Xi^{1/4}$, and summing such estimates brings the logarithm back. As a consequence of Theorem 2, $C_\varphi \in S_p$ if and only if $p > 2/(d-1)$, which was already known from (2). The lower bound in [3, Theorem 1.2(b)] on $\mathscr{H}^p$ also carries the factor $\log n$; the proof of Theorem 1 uses orthogonality and does not apply there.

**General critical symbols.** Bayart and Brevig [1, Theorem 16] prove, for symbols with a unique boundary-regular contact point, that $(1/n)^{\eta_\varphi} \ll a_n(C_\varphi) \ll (\log n/n)^{\eta_\varphi}$. We have not examined whether the dyadic splitting of Section 5 applies in that generality.

**Problem 1.** *Let $\varphi$ be as in* [1, Theorem 16]. *Is it true that $a_n(C_\varphi) \asymp n^{-\eta_\varphi}$?*

**The best uniform gain.** Theorems 3 and 4 show that the best $g_d$ such that every symbol as in Theorem 3 maps $\mathscr{D}_\alpha$ boundedly into $\mathscr{D}_\beta$ whenever $d\beta + 1 + g_d \ge 2^\alpha$ satisfies $1/8 \le g_d \le (d-1)/4$. The value $1/8$ comes from the crude bound $m \le 8$ in case (c) of the proof, which treats every two-variable restriction separately and does not use the other variables.

**Problem 2.** *Let $d \ge 3$. Determine the largest $g_d$ such that every Dirichlet polynomial $\varphi \in \mathscr{G}$ with unrestricted range,* $\dim(\varphi) = d$ *and* $\deg(\varphi) \in \{1, 2\}$ *satisfies the following: $C_\varphi$ maps $\mathscr{D}_\alpha$ boundedly into $\mathscr{D}_\beta$ for all $\alpha > 0$ and $\beta \ge 0$ with $d\beta + 1 + g_d \ge 2^\alpha$.*

**Code availability.** The script which verifies the identities in the proof of Lemma 7 requires SymPy and runs in about a second. It computes the Taylor coefficients of $f_1$ and $f_3$ over the field $\mathbb{Q}(a_1, a_2, x_1, x_2, \omega)$ and checks the relations modulo $S^3\omega^2 + V$ by polynomial division, so no floating point arithmetic is involved. It is available at `https://github.com/Oisov/math-preprints`, in the directory `numerics/composition-sharpness/`.

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University of Oslo, Oslo, Norway
*Email address*: oistein.sovik@usit.uio.no